Supplemental document

Kernel learning enables fluorescence microscopic image deconvolution with enhanced performance and speed

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Section S1. The convergence of different backward kernels

For the noise-free data, all the methods showed a convergent trend, and KLD demonstrated a faster convergence compared to other methods (Fig. S2A). However, for the noisy data, all the methods first had a convergent trend and then diverged whether or not it used a matched backward kernel (Fig. S2B). Using unmatched backward kernels (KLD, WB, Butterworth, and Gaussian), algorithms diverged faster from the desired solution compared to that using a matched kernel (Traditional). A more efficient backward kernel gives a larger update step for each iteration and therefore results in a faster convergence and divergence rate. Although KLD showed fastest divergent, it gave a faster convergent rate in the "convergent period" and obtained a better solution with the number of iterations used at training stage (here, 2 iterations). Therefore, setting the number of iterations used at prediction stage to be the same as that at training stage can avoid the risk of algorithm divergence.

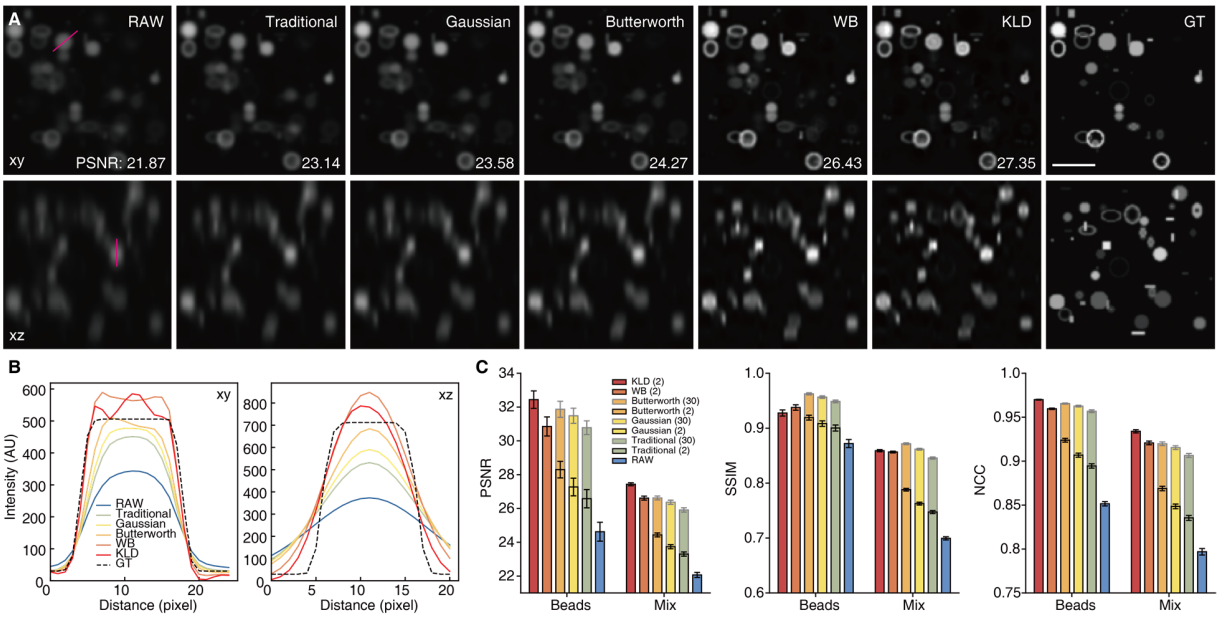


Fig. S1. Performance of KLD on noise-free 3D simulated images. (A) Lateral (xy) and axial (xz) views of simulated noisy images with mixed structures, comparing raw low-resolution input (RAW), GT, RLD with traditional, Gaussian, Butterworth and WB backward kernels, and KLD. The PSNR of each volume is shown at the bottom right of its lateral view. KLD was trained on noise-free simulated data with mixed structures. The iteration number of all the methods were set as 2. (B) Intensity profiles along the magenta lines in the xy planes (left) and xz planes (right) in (A), showing the result of KLD is closer to the ground truth than other methods. (C) Quantitative analysis (mean ± standard deviation, n = 6 volumes for data with bead structures and data with mixed structures) with PSNR, SSIM, and NCC for the predictions of different methods, demonstrating the good generalization and better performance of KLD. The number of iterations was shown in the parenthesis after each method in the legend. KLD was trained using data with mixed structures. Scale bar, 5 μm (A).

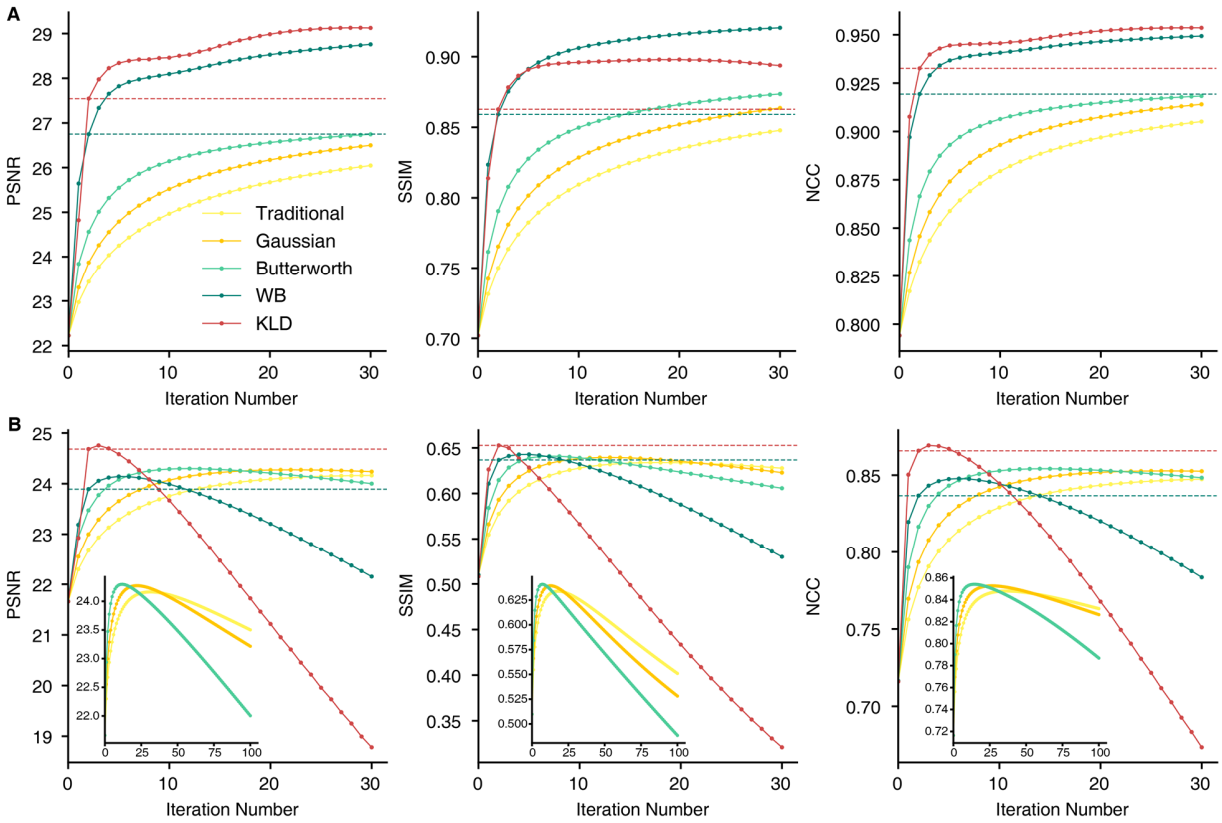


Fig. S2. The convergence of KLD and RLD using different backward kernels. The PSNR, SSIM and NCC of a simulated noise-free (A) and a noisy (B) sample with mixed structures after deconvolution as a function of iteration number compared for KLD and RLD using traditional, Gaussian, Butterworth and WB backward kernels. The iteration number of KLD was set as 2 at training stage. The red and green dash lines denote the metric value of the result of KLD and RLD with WB kernel at the second iteration, respectively.

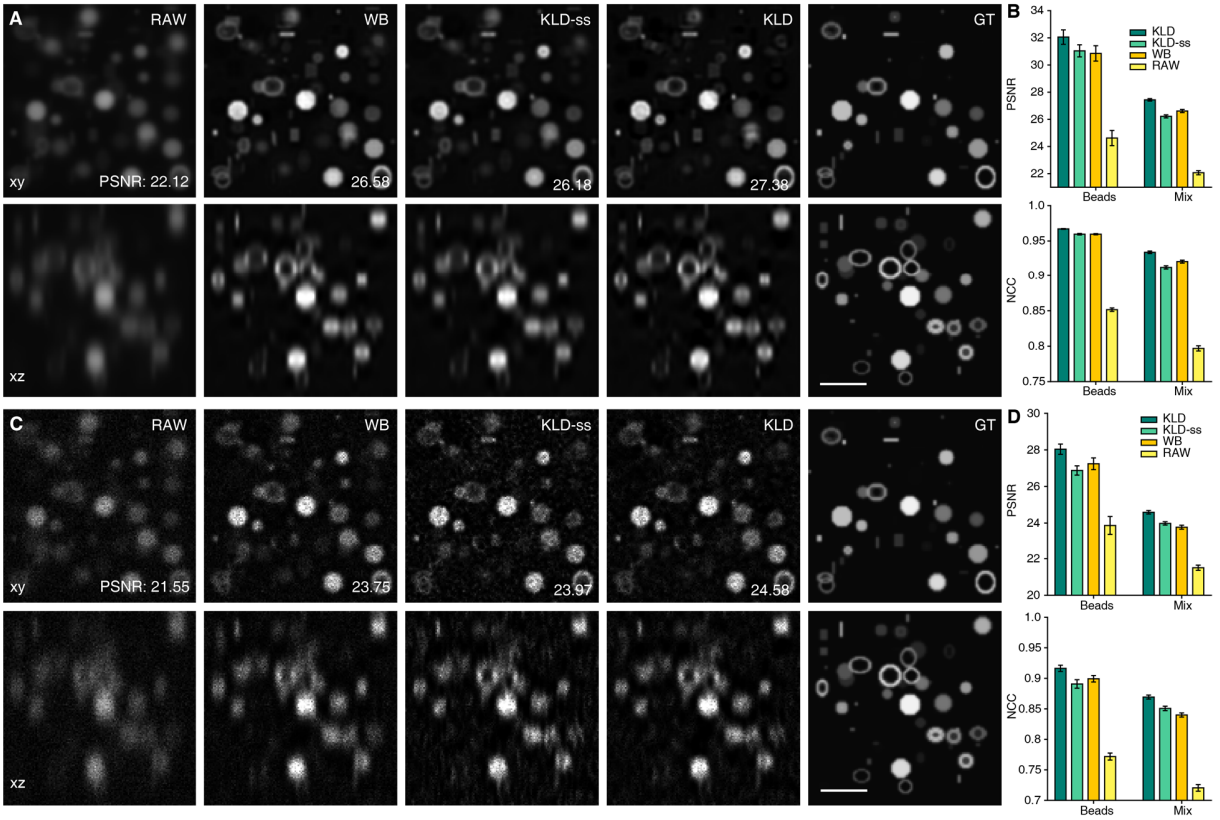


Fig. S3. The performance of KLD trained using self-supervised training strategy on simulation data set. (A and C), Lateral (xy) and axial (xz) views of a simulated noise-free (A) and a noisy (C) image with mixed structures, comparing raw low-resolution input (RAW), GT, RLD with WB backward kernel, KLD using self-supervised training strategy (KLD-ss) and KLD using supervised training strategy (KLD). The PSNR of each volume is shown at the bottom right of its lateral view. KLD was trained on simulated data set with mixed structures. The iteration number of all the methods were set as 2. (B and D) Quantitative analysis (mean ± standard deviation, n = 6 volumes for data with bead structures and data with mixed structures) with PSNR and NCC for the predictions of different methods on noise-free (B) and noisy (D) data. Scale bar, 5 μm [(A) and (C)].

Table S1. Parameters used in the training of KLD.

Data set	**Simulation**		**BioSR**		**Confocal/STED volume**		**LLSM volume**	
Microscope	iSPIM		WF/SIM		Confocal/STED		LLSM	
Kernel	k_f	k_b	k_f	k_b	k_f	k_b	k_b (simulation)	k_b (self-supervise)
Size of kernel	31×31×31		31×31		31×31×3		101×101×101	
Training volumes	1/2/3		1		1		1	1000
Training volume size	128×128×128		502×502		1024×1024×6		382x382x151	
Learning rate	0.01	1×10^{-6}	0.001	1×10^{-5}	0.01	1×10^{-5}	1×10^{-7}	1×10^{-7}
Decay step	-	2000	-	2000	-	2000	2000	2000
Decay rate	-	0.5	-	0.5	-	0.5	-	-
Training iteration number	500	10000	500	10000	500	10000	4000	3000
Training time	~10 min	~8 h	~1 min	~ 40 min	~ 15 min	~5 h	~16h	~12h
Processing time for each image	~0.2s		~0.03s		~1s		~4s	

Table S2. Parameters used in RLD with traditional, Gaussian, Butterworth and WB kernels, and DeconvBlind.

		Traditional	Gaussian	Butterworth			WB				DeconvBlind
Data set	**Specimen**	**Iteration number**	**Iteration number**	**beta**	**n**	**Iteration number**	**alpha**	**beta**	**n**	**Iteration number**	**Iteration number**
Simulation (noise-free)	**Mix**	30	30	0.01	10	30	0.005	0.1	10	2	-
	Beads	30	30	0.01	10	30	0.005	0.1	10	2	-
Simulation (noisy)	**Mix**	30	30	0.01	10	30	0.005	0.001	10	2	-
	Beads	30	30	0.01	10	30	0.005	0.001	10	2	-
BioSR	**MT**	100	-	-	-	-	-	-	-	-	6
	F-actin	100	-	-	-	-	-	-	-	-	8
Confocal/STED volume	**MT**	20	-	-	-	-	-	-	-	-	3
	NPC	20	-	-	-	-	-	-	-	-	3
LLSM volume	**Mito**	30	30	0.01	10	30	0.005	0.1	10	2	-
	H2B	30	30	0.01	10	30	0.005	0.1	10	2	-

Movie S1. Subcellular dynamics of Mito (hot) and H2B (gray) during mitosis. Two-color MIP images of raw input (RAW) and images deconvolved by RLD with WB kernel and KLD for 250 time points at 40 second intervals in a HeLa cell. The slice numbers of MIP = 10. Scale bar, 5 μm.

Movie S2. Instructional video for the KLD napari plugin. Screen captures of the installation, training and inference procedures of the KLD napari plugin, as well as the process of generating simulation data.